

Geometry
Unit 6
Lesson 9 Homework

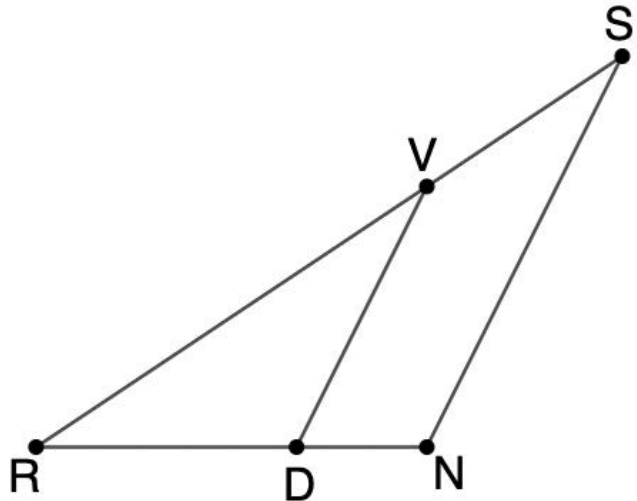
Name Answer Key
Date _____
Period _____

$\triangle RSN$ with $\overline{VD}$ is shown.

Given: $RV = \frac{2}{3}RS$ and
 $RD = \frac{2}{3}RN$

Prove: $\triangle RSN \sim \triangle RVD$

Complete the two-column proof with the provided statements and reasons.



Statement	Reason
$RV = \frac{2}{3}RS$ and $RD = \frac{2}{3}RN$	Given
1. $\frac{RV}{RS} = \frac{2}{3}$ and $\frac{RD}{RN} = \frac{2}{3}$	Division Property of Equality
2. $\frac{RV}{RS} = \frac{RD}{RN}$	Transitive Property of Equality
3. $\angle VRD \cong \angle SRN$	Reflexive Property of Congruence
4. $\triangle RSN \sim \triangle RVD$	SAS ~

$\triangle RSN \sim \triangle RVD$

SAS ~

$\frac{RV}{RS} = \frac{RD}{RN}$

Transitive Property of Equality

$\angle VRD \cong \angle SRN$

Division Property of Equality

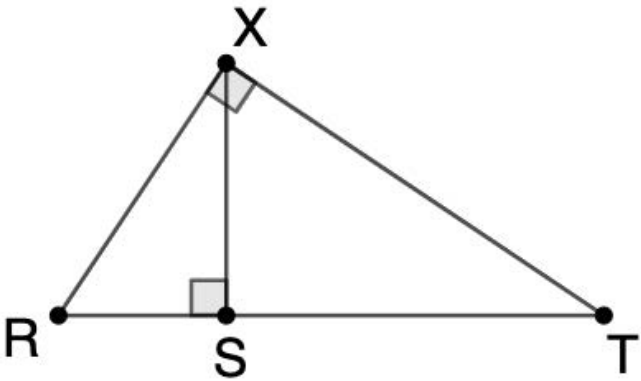
$\frac{RV}{RS} = \frac{2}{3}$ and $\frac{RD}{RN} = \frac{2}{3}$

Reflexive Property of Congruence

$\triangle RXT$ with $\overline{XS}$ is shown.

Given: $\overline{XS} \perp \overline{RT}$ and
 $\overline{RX} \perp \overline{XT}$

Prove: $\triangle XTR \sim \triangle SXR$



Complete the two-column proof with the provided statements and reasons.

	Statement	Reason
	$\overline{XS} \perp \overline{RT}$ and $\overline{RX} \perp \overline{XT}$	Given
5.	$\angle XSR$ and $\angle TXR$ are right angles	Definition of perpendicular lines
6.	$m\angle XSR = 90^\circ$ and $m\angle TXR = 90^\circ$	Definition of right angles
7.	$m\angle XSR = m\angle TXR$	Transitive Property of Equality
8.	$\angle XSR \cong \angle TXR$	Definition of congruence
9.	$\angle XRS \cong \angle TRX$	Reflexive Property of Congruence
10.	$\triangle XTR \sim \triangle SXR$	AA ~

$\angle XSR \cong \angle TXR$	AA ~
$\triangle XTR \sim \triangle SXR$	Definition of congruence
$\angle XRS \cong \angle TRX$	Definition of perpendicular lines
$m\angle XSR = m\angle TXR$	Reflexive Property of Congruence
$m\angle XSR = 90^\circ$ and $m\angle TXR = 90^\circ$	Transitive Property of Equality
$\angle XSR$ and $\angle TXR$ are right angles	Definition of right angles